AC9M6N03 • Year 6 Mathematics

Equivalent Fractions on a Common Number Line

Compare halves, thirds, quarters and related fractions and justify their order

We are learning to…

Students rename fractions with common denominators, use benchmarks and place proper, improper and mixed forms accurately on one continuous number line.

Represent | Reason | Calculate | Interpret

Verify

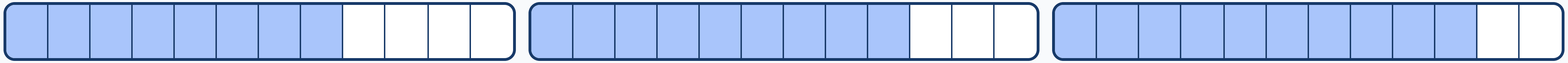
Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

apply knowledge of equivalence to compare, order and represent common fractions including halves, thirds and quarters on the same number line and justify their order

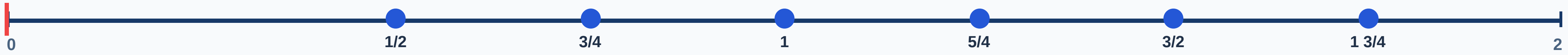
Compare 2/3, 3/4 and 5/6



2/3 = 8/12 **3/4 = 9/12** **5/6 = 10/12**

A common denominator creates equal-sized parts, so numerators can be compared meaningfully. Equivalent forms occupy the same point.

Locate fractions greater than one



Partition each whole into equal intervals and preserve the denominator across whole-number boundaries.

Build and annotate the model

Represent equivalent fractions on a common number line and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Larger denominator means larger value:** For the same numerator, more parts means smaller parts.
- **Different wholes compared:** Fractions must refer to equal-sized wholes.
- **Improper fraction placed below one:** Compare numerator with denominator first.
- **Number-line intervals drawn unequally:** Equal numerical steps require equal spacing.

Quick check

- Order 2/3, 3/4 and 5/6.
- Show 3/4 = 9/12.
- Locate 5/4.
- Compare 7/8 and 5/6.
- Use 1/2 as a benchmark.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.